

Lecture 12: Directional Derivatives & the Gradient

Background & Motivation

Partial derivatives measure change along the coordinate axes. But what if we want the rate of change in an arbitrary direction? The directional derivative answers this, and the gradient vector ∇f tells us the direction of fastest increase. The gradient is one of the most important objects in all of applied mathematics—it drives optimization, machine learning, physics, and engineering.

1. Notation Review

- We write vectors in bold: $\mathbf{x} = \langle x_1, x_2 \rangle$ or $\mathbf{x} = \langle x_1, x_2, x_3 \rangle$
- A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ takes a vector input and returns a scalar
- Partial derivatives: f_x , f_y , or $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$

2. The Directional Derivative

Definition 1: Directional Derivative (Limit Definition)

The **directional derivative** of f at $\mathbf{p}$ in the direction of unit vector $\hat{\mathbf{u}}$:

$$D_{\hat{\mathbf{u}}}f(\mathbf{p}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{p} + h\hat{\mathbf{u}}) - f(\mathbf{p})}{h}$$

- Measures the rate of change of f in the direction $\hat{\mathbf{u}}$
- Special cases: $D_{\hat{\mathbf{i}}}f = f_x$ and $D_{\hat{\mathbf{j}}}f = f_y$
- **$\hat{\mathbf{u}}$ must be a unit vector**

3. The Gradient Vector

Definition 2: Gradient

The **gradient** of $f(x, y)$ – the column of all partial derivatives:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

For $f(x, y, z)$:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

Theorem 1: Directional Derivative via the Gradient

If f is differentiable, then:

$$D_{\hat{\mathbf{u}}}f(\mathbf{p}) = \nabla f(\mathbf{p}) \cdot \hat{\mathbf{u}}$$

Example 1: Computing the Gradient at a Point

Let $f(x, y, z) = x^2y + yz^3 - \sin(xz)$. Find ∇f at $(1, 2, 0)$ and the directional derivative in the direction $\langle 1, -1, 2 \rangle$.

1. **Compute partials:**

$$f_x = 2xy - z \cos(xz), \quad f_y = x^2 + z^3, \quad f_z = 3yz^2 - x \cos(xz)$$

2. **Evaluate at $(1, 2, 0)$:**

$$\nabla f(1, 2, 0) = \langle 4, 1, -1 \rangle$$

3. **Get unit vector:** $\hat{\mathbf{u}} = \frac{1}{\sqrt{6}} \langle 1, -1, 2 \rangle$

4. **Compute:** $D_{\hat{\mathbf{u}}}f = \langle 4, 1, -1 \rangle \cdot \frac{1}{\sqrt{6}} \langle 1, -1, 2 \rangle = \frac{4 - 1 - 2}{\sqrt{6}} = \frac{1}{\sqrt{6}}$

Example 2: Verification: Gradient Formula vs. Limit Definition

For $f(x, y) = x^2 + 3xy + y^2$ at $(1, 2)$ in direction $\hat{\mathbf{u}} = \frac{1}{\sqrt{2}} \langle 1, 1 \rangle$, compute $D_{\hat{\mathbf{u}}}f$ both ways.

Method 1 – Gradient formula:

1. **Gradient:** $\nabla f = \langle 2x + 3y, 3x + 2y \rangle$

2. **Evaluate:** $\nabla f(1, 2) = \langle 8, 7 \rangle$

3. **Dot product:** $D_{\hat{\mathbf{u}}}f = \langle 8, 7 \rangle \cdot \frac{1}{\sqrt{2}} \langle 1, 1 \rangle = \frac{15}{\sqrt{2}}$

Method 2 – Limit definition:

1. **Set up:** $D_{\hat{\mathbf{u}}}f = \lim_{h \rightarrow 0} \frac{f\left(1 + \frac{h}{\sqrt{2}}, 2 + \frac{h}{\sqrt{2}}\right) - f(1, 2)}{h}$

2. **Expand:** $f\left(1 + \frac{h}{\sqrt{2}}, 2 + \frac{h}{\sqrt{2}}\right) = \left(1 + \frac{h}{\sqrt{2}}\right)^2 + 3\left(1 + \frac{h}{\sqrt{2}}\right)\left(2 + \frac{h}{\sqrt{2}}\right) + \left(2 + \frac{h}{\sqrt{2}}\right)^2$

After expanding and subtracting $f(1, 2) = 12$: numerator = $\frac{15h}{\sqrt{2}} + O(h^2)$

3. **Result:** $D_{\hat{\mathbf{u}}}f = \lim_{h \rightarrow 0} \frac{15h/\sqrt{2}}{h} = \frac{15}{\sqrt{2}} \quad \checkmark \quad \text{Same answer!}$

4. Key Properties of the Gradient

Theorem 2: Gradient Properties

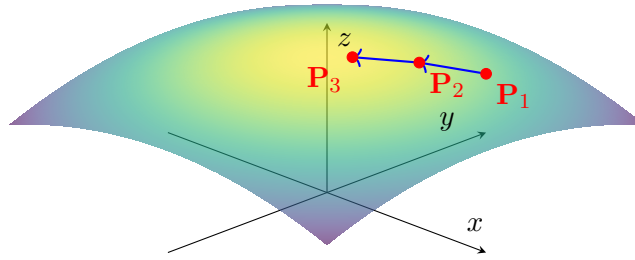
Let f be differentiable at $\mathbf{p}$ with $\nabla f(\mathbf{p}) \neq \mathbf{0}$. Then:

1. ∇f points in the direction of **steepest increase (ascent)**
2. $-\nabla f$ points in the direction of **steepest decrease (descent)**

3. $\|\nabla f\|$ equals the **maximum rate of change of f**
4. ∇f is **perpendicular to level curves/surfaces** of f
5. $D_{\hat{\mathbf{u}}}f = \|\nabla f\| \cos \theta$ where θ is the angle between ∇f and $\hat{\mathbf{u}}$

– **Why perpendicular to level curves?** If $\mathbf{r}(t)$ lies on $f = k$, then $\frac{d}{dt}[f(\mathbf{r}(t))] = \nabla f \cdot \mathbf{r}'(t) = 0$. Since $\mathbf{r}'(t)$ is tangent to the level curve, $\nabla f \perp$ level curve.

Gradient Ascent: Local Surface and Gradient Vectors



Gradient ascent on $f(x, y) = 4 - x^2 - y^2$. Starting at $\mathbf{P}_1$, each step moves in the direction of $+\nabla f$, climbing toward the maximum at the origin. Gradient descent (to minimize) uses $-\nabla f$ instead.

5. Gradient Ascent and Descent

Gradient Descent Algorithm

To minimize $f(\mathbf{x})$:

1. **Compute** $\nabla f(\mathbf{p}_n)$
2. **Unit direction:** $\hat{\mathbf{d}} = -\frac{\nabla f}{\|\nabla f\|}$ (steepest decrease)
3. **Update:** $\mathbf{p}_{n+1} = \mathbf{p}_n + \eta \hat{\mathbf{d}} = \mathbf{p}_n - \eta \frac{\nabla f(\mathbf{p}_n)}{\|\nabla f(\mathbf{p}_n)\|}$
4. **Iterate** until convergence

For **gradient ascent** (maximize), use $+\nabla f$ instead.

Example 3: Gradient Descent on $f = x^2 + 4y^2$

Minimize $f(x, y) = x^2 + 4y^2$ starting from $\mathbf{p}_0 = (4, 2)$ with step size $\eta = 0.5$. Do 3 iterations.

$$\nabla f = \langle 2x, 8y \rangle$$

$$\textbf{Iteration 1: } \nabla f(4, 2) = \langle 8, 16 \rangle, \quad \|\nabla f\| = \sqrt{64 + 256} = 8\sqrt{5}$$

$$\mathbf{p}_1 = (4, 2) - \frac{0.5}{8\sqrt{5}} \langle 8, 16 \rangle = (4, 2) - \langle 0.224, 0.447 \rangle \approx (3.776, 1.553)$$

$$f(\mathbf{p}_1) \approx 14.26 + 9.65 = 23.9 \quad (\text{down from } f(\mathbf{p}_0) = 32)$$

$$\textbf{Iteration 2: } \nabla f(3.776, 1.553) \approx \langle 7.55, 12.42 \rangle, \quad \|\nabla f\| \approx 14.53$$

$$\mathbf{p}_2 \approx (3.776, 1.553) - \frac{0.5}{14.53} \langle 7.55, 12.42 \rangle \approx (3.516, 1.126)$$

$$f(\mathbf{p}_2) \approx 12.36 + 5.07 = 17.4$$

$$\textbf{Iteration 3: } \nabla f(3.516, 1.126) \approx \langle 7.03, 9.01 \rangle, \quad \|\nabla f\| \approx 11.43$$

$$\mathbf{p}_3 \approx (3.516, 1.126) - \frac{0.5}{11.43} \langle 7.03, 9.01 \rangle \approx (3.209, 0.732)$$

$$f(\mathbf{p}_3) \approx 10.30 + 2.14 = 12.4$$

Each iteration decreases f : $32 \rightarrow 23.9 \rightarrow 17.4 \rightarrow 12.4$, converging toward the minimum at $(0, 0)$.

Example 4: Gradient Ascent on $f = -x^2 - y^2 + 4x + 4y$

Maximize $f(x, y) = -x^2 - y^2 + 4x + 4y$ starting from $\mathbf{p}_0 = (0, 0)$ with $\eta = 0.5$. Do 3 iterations.

$$\nabla f = \langle -2x + 4, -2y + 4 \rangle$$

Iteration 1: $\nabla f(0, 0) = \langle 4, 4 \rangle$, $\|\nabla f\| = 4\sqrt{2}$

$$\mathbf{p}_1 = (0, 0) + \frac{0.5}{4\sqrt{2}} \langle 4, 4 \rangle = \left(\frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}} \right) \approx (0.354, 0.354)$$

$$f(\mathbf{p}_1) \approx -0.125 - 0.125 + 1.414 + 1.414 = 2.58 \quad (\text{up from } f(\mathbf{p}_0) = 0)$$

Iteration 2: $\nabla f(0.354, 0.354) = \langle 3.29, 3.29 \rangle$, $\|\nabla f\| \approx 4.65$

$$\mathbf{p}_2 \approx (0.354, 0.354) + \frac{0.5}{4.65} \langle 3.29, 3.29 \rangle \approx (0.708, 0.708)$$

$$f(\mathbf{p}_2) \approx -0.501 - 0.501 + 2.832 + 2.832 = 4.66$$

Iteration 3: $\nabla f(0.708, 0.708) = \langle 2.58, 2.58 \rangle$, $\|\nabla f\| \approx 3.65$

$$\mathbf{p}_3 \approx (0.708, 0.708) + \frac{0.5}{3.65} \langle 2.58, 2.58 \rangle \approx (1.061, 1.061)$$

$$f(\mathbf{p}_3) \approx 5.88$$

Converging toward the maximum at $(2, 2)$ where $f(2, 2) = 8$: $0 \rightarrow 2.58 \rightarrow 4.66 \rightarrow 5.88$.

Connection to Machine Learning

- In ML, a model has parameters $\mathbf{w}$ and a **loss function** $L(\mathbf{w})$ measuring error
- Training is gradient descent on L : $\mathbf{w}_{n+1} = \mathbf{w}_n - \eta \nabla L(\mathbf{w}_n)$
- The step size η is called the **learning rate**
- This is how neural networks, logistic regression, and most modern AI models are trained

Practice Problems

Problem 1. Find ∇f for $f(x, y) = xe^y + \sin(xy)$ at the point $(0, \pi)$.

$$\nabla f = \langle e^y + y \cos(xy), xe^y + x \cos(xy) \rangle. \text{ At } (0, \pi): \nabla f = \langle e^\pi + \pi, 0 \rangle.$$

Answer: $\nabla f(0, \pi) = \langle e^\pi + \pi, 0 \rangle$

Problem 2. Find the maximum rate of change of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at $(3, 4, 12)$ and the direction in which it occurs.

$$\nabla f = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \langle x, y, z \rangle. \text{ At } (3, 4, 12): f = 13.$$

$$\nabla f = \frac{1}{13} \langle 3, 4, 12 \rangle. \text{ Max rate} = \|\nabla f\| = 1. \text{ Direction: } \frac{1}{13} \langle 3, 4, 12 \rangle.$$

Problem 3. Find the directional derivative of $f(x, y) = x^2 - 3y^2$ at $(2, -1)$ in the direction toward the point $(5, 3)$.

Direction: $\langle 5 - 2, 3 - (-1) \rangle = \langle 3, 4 \rangle$, $\hat{\mathbf{u}} = \langle 3/5, 4/5 \rangle$.

$\nabla f = \langle 2x, -6y \rangle$. At $(2, -1)$: $\nabla f = \langle 4, 6 \rangle$.

$D_{\hat{\mathbf{u}}}f = \langle 4, 6 \rangle \cdot \langle 3/5, 4/5 \rangle = 12/5 + 24/5 = 36/5$.

Problem 4. Perform 2 iterations of gradient descent on $f(x, y) = (x - 3)^2 + (y + 1)^2$ from $\mathbf{p}_0 = (0, 0)$ with $\eta = 1$.

$\nabla f = \langle 2(x - 3), 2(y + 1) \rangle$

Iteration 1: $\nabla f(0, 0) = \langle -6, 2 \rangle$, $\|\nabla f\| = \sqrt{40} = 2\sqrt{10}$.

$\mathbf{p}_1 = (0, 0) - \frac{1}{2\sqrt{10}} \langle -6, 2 \rangle = \left(\frac{3}{\sqrt{10}}, -\frac{1}{\sqrt{10}} \right) \approx (0.949, -0.316)$

$f(\mathbf{p}_1) \approx (0.949 - 3)^2 + (-0.316 + 1)^2 = 4.21 + 0.47 = 4.68$ (down from 10)

Iteration 2: $\nabla f(0.949, -0.316) = \langle -4.10, 1.37 \rangle$, $\|\nabla f\| \approx 4.32$

$\mathbf{p}_2 \approx (0.949, -0.316) - \frac{1}{4.32} \langle -4.10, 1.37 \rangle \approx (1.898, -0.633)$

$f(\mathbf{p}_2) \approx 1.21 + 0.13 = 1.35$ (converging toward min at $(3, -1)$)

Answer: $\mathbf{p}_2 \approx (1.90, -0.63)$

Problem 5. Show that ∇f is perpendicular to level curves. *Hint:* If $\mathbf{r}(t)$ is a curve on the level surface $f = k$, differentiate $f(\mathbf{r}(t)) = k$.

$$\frac{d}{dt}[f(\mathbf{r}(t))] = \frac{d}{dt}[k] = 0$$

By the chain rule: $\nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 0$

Since $\mathbf{r}'(t)$ is tangent to the level curve, this says $\nabla f \perp$ the level curve. $\square$